A Parsimonious Model for Forecasting Gross Box-Office Revenues of Motion Pictures

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Abstract

The primary objective of this paper is to develop a parsimonious model for forecasting the gross box-office revenues of new motion pictures based on early box office data. The paper also seeks to provide insights into the impact of distribution policies on the adoption of new products. The model is intended to assist motion picture exhibitor chains (retailers) in managing their exhibition capacity and in negotiating exhibition license agreements with distributors (studios), by allowing them to project the box-office potential of the movies they plan to or currently exhibit based on early box-office results. It is also of interest to practitioners in other software industries (e.g., music, books, CD-ROMs) where the distribution intensity is highly variable over the product life cycle and is an important determinant of new product adoption patterns. The model and its extensions are of interest to academic researchers interested in modeling distribution effects in new product adoption, as well as forecasters looking for ways to leverage historical data on related products to forecast the sales of new products.

We draw upon a queuing theory framework to conceptualize stochastically the consumer's movie adoption process in two steps—the *time to decide* to see the new movie, and the *time to act* on the adoption decision. The parameter for the time-to-decide process captures the intensity of information intensity flowing from various information sources, while the parameter for the time-to-act process is related to the delay created by limited distribution intensity and other factors. Our conceptualization extends existing new product forecasting models, which assume that consumers act instantaneously on the motivating information they receive about the new product. The resulting model is parsimonious, yet it accommodates a wide range of adoption patterns. In addition, the stochastic formulation allows us to quantify the uncertainty surrounding the expected adoption pattern. In the empirical testing, we focus on the most parsimonious version of the modeling framework, BOXMOD-I, a model that assumes stationarity with respect to the two shape parameters that characterize the adoption process. The model produces fairly accurate early forecasts using at most the first three weeks of data for calibration, and the predictive performance of the model compares favorably with benchmark models. We propose extensions of the basic model that account for more realistic nonstationary distribution intensity patterns—including a "wide release" pattern that relies on intensive distribution and promotion, and a "platform release" pattern that involves a gradual buildup of distribution intensity. Finally, we present an adaptive weighing scheme that combines initial parameter estimates obtained from a meta-analysis procedure with estimates obtained from early data to produce forecasts of box-office revenues for a new movie when little or no box-office data are available.

An important finding from the empirical testing is that motion picture box-office revenue patterns display remarkable empirical regularity. We find that there are only three classes of adoption patterns, and these can all be represented within the basic model by using a two-parameter Exponential or Erlang-2 probability distribution, or a three parameter Generalized Gamma distribution. We also find that cumulative box-office revenues can be predicted with reasonable accuracy (often within 10% of the actual) using as little as two or three data points. However, our attempts to predict revenue patterns without any sales data meet with limited success. While the scale parameter can be estimated reasonably well from a historical database of parameter values, we find that it is considerably more difficult to predict the shape parameters using historical data.

The parsimony we seek in developing the model comes at the cost of several limiting assumptions. We assume that the time-to-decide subprocess and the time-to-act subprocess are independent, which may not be the case if decisions on continued exhibition by retailers are endogenously related to box-office revenues over the life cycle. In the basic model formulation, we also assume that the time-to-act process can be represented by an exponential distribution, which may not always be the case. While we provide some empirical evidence to support these assumptions, further research could relax these and other assumptions to enrich the basic model, although this would entail some loss in parsimony.

(*Forecasting; Motion Pictures; Distribution; Stochastic Models; Consumer Behavior*)

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1. Introduction

In this paper, we address the problem of early forecasting of the performance of new entertainment "software" products such as motion pictures, taking the perspective of the retailers (called "exhibitors" in the motion picture industry) of such products. There were 26,689 screens or "outlets" for motion pictures in the United States at the end of 1994. The motion picture exhibition industry is dominated by chains such as United Artists with 2,300 screens, Carmike Cinemas with 2,190 screens, and Cineplex Odeon with 1,617 screens (Deutsch 1995). These exhibitor chains need to manage the yield from their exhibition capacity, based on their estimates of demand for the movies that they are currently exhibiting. Unlike physical products, where it is difficult to change distribution intensity at short notice due to inventory and logistical constraints, it is easy for movie exhibitors to adapt the exhibition capacity allocated to a new movie from week to week, either by dropping the movie from the multiplex theater or by shifting it to a smaller screening room. Exhibitors of movies are thus very much interested in early forecasts of the gross box-office revenues of a movie in making their exhibition decisions.

We focus on the motion picture industry, since this industry has some unique characteristics that make it very interesting to research from the perspective of new product forecasting. Most of the revenues in the motion picture industry are derived from new movies, since the shelf life of a typical movie is less than 15 weeks in the domestic theatrical release. Hundreds of new movies are launched every year. Demand for a new movie is highly uncertain, since movies are "experiential" products, and it is difficult for consumers to evaluate the quality of a movie until they have actually experienced it (Hirschman and Holbrook 1982, Holbrook and Hirschman 1982, Eliashberg and Sawhney 1994). Given the importance of new movies and the uncertainty in predicting the box-office (B.O.) performance of these new movies, the value of accurate box-office forecasts is extremely high in this industry.

Our primary objective is to provide movie exhibitor chains like Cineplex Odeon and United Artists with a relatively simple and accurate forecasting tool that can assist them in maximizing the yield from their exhibition capacity in multiplex theaters. Our model could assist exhibitor chains in making exhibition decisions beyond the initial contractual periods of three to five weeks, which are based on week-to-week negotiations between exhibitors and distributors as the box-office performance of a new movie unfolds (Squire 1992, De Vany and Eckert 1991).

To make decisions on continued exhibition of new movies beyond the contractual period, exhibitor chains need to forecast the weekly total contribution for each movie on a *per screen* basis from two sources: contribution from ticket sales, and contribution from concession sales (Durwood and Rutkowski 1992). The exhibitor's contribution from ticket sales in a specific week depends on the B.O. revenue per screen for the week, and the contractual share of the B.O. grosses that accrue to the exhibitor. B.O. grosses are shared between distributors and exhibitors on a sliding scale that typically begins at 90%–10% in favor of the distributor, and progresses to 50%–50% several weeks into the life cycle (Friedberg 1992). This is why exhibitors, as opposed to distributors, are concerned with revenues from the later stages of the movie's life cycle. Contribution from concession sales is approximately 75% of concession sales, which in turn range from 10% to 20% of gross Box-Office revenues (Squire 1992). Therefore, if exhibitors can forecast the weekly per-screen B.O. revenues for their chain, they can easily make decisions on continued exhibition of new movies.

In developing our forecasting model, we generate estimates for the *national* B.O. revenues of a new movie. But the exhibitor chain needs to know the expected box-office revenues per screen for *their individual screens*. Since a major exhibitor chain accounts for a large portion of the approximately 2,000 screens at which a major motion picture typically opens, the expected national per-screen revenues are fairly representative of the exhibitor chain's expected per-screen B.O. revenues for the movie. These per-screen B.O. estimates can then be used to forecast the total contribution from ticket and concession sales. With this understanding, we will focus on estimating national box-office revenues, having shown how these forecasts could be used by an exhibitor chain to make decisions at the individual screen level.

Since our focus is on prediction and ease of use, we adopt a position of parsimony in developing the modeling framework. The basic model we develop and test has only three parameters and therefore requires only three weeks of data to generate forecasts. The model is easy to understand, and it can be estimated using any off-the-shelf desktop statistical software package. Our emphasis on parsimony necessarily comes at the cost of making several simplifying assumptions. We provide support for some of these modeling assumptions. We also demonstrate how some of the underlying assumptions can be relaxed by presenting extensions of our modeling framework to incorporate more flexible functional forms and explanatory variables. However, we emphasize the parsimonious version of the model in our empirical testing.

In comparing our modeling framework against other new product forecasting models, the models of Jones and Ritz (1991) and Jo-nes and Mason (1990) are very directly relevant. These models have been inspired by the deterministic Bass (1969) first purchase diffusion model. Jones and Ritz (1991) incorporate distribution (exhibition) effects into the new motion picture diffusion process by conceptualizing the process in terms of two parallel subprocesses: the *retailer* (screens) diffusion process and the *consumer* diffusion process. They link the retailer diffusion process to the consumer diffusion process by assuming that each retailer who adopts the new product increases the pool of potential consumers by a fixed amount. This yields a four-parameter model which is tested by the authors for its goodness-of-fit, based on the entire data set, rather than predictive results based on a relatively small number of data points. In a related model, Jones and Mason (1990) extend their previous model and propose a more elaborate model that can accommodate additional distribution-related phenomena. However, the model has a large number of parameters, making it very cumbersome to estimate and use. The authors do not discuss empirical estimation of their model, and resort to numerical simulations to illustrate the model's properties. Given the number of model parameters and the lack of guidelines for parameter estimation, the utility of the Jones and Mason (1990) model for early sales forecasting seems quite limited.

In this paper, we develop and test a more parsimonious model that provides reasonably accurate forecasts with as few as three data points. Further, while the other two models are deterministic and therefore offer no insights into the uncertainty surrounding the consumer adoption pattern, our model employs a stochastic framework, which allows us to estimate the dynamic uncertainty (variance) in the adoption pattern in addition to the expected value (mean) of the pattern. In the motion picture industry, which is characterized by a high degree of unpredictability, the quantification of uncertainty is a useful insight.

2. Modeling Framework

2.1. Conceptualization of the Moviegoing Process and Mathematical Development

We conceptualize an individual's adoption of a new movie as a two-step process. This conceptualization is inspired by previous modeling efforts on the spread of a motivating piece of information from a source (Berg 1981). While Berg (1981, p. 142) refers to the source as "some media," conceptually there is no reason to exclude other sources of motivating information (e.g., critics' favorable reviews, positive word-of-mouth). In the first step, the individual is exposed to the influential information about the new movie. In the second step, the individual waits for some time before acting upon the decision to see the movie. Thus, we model the individual's (stochastic) time to adopt (see) a new movie as the sum of the (stochastic) *time to decide* to see the new movie; and the (stochastic) *time to act* on the decision. The explicit recognition of the time to act represents an extension of previous modeling work which assumes that individual acts immediately upon receiving a motivating piece of information (Chatterjee and Eliashberg 1990, Roberts and Urban 1988).

Technically, we follow Berg (1981) and model the time to decide and the time to act as two independent stochastic processes. The assumption of independence implies that the time that it takes an individual to hear a motivating piece of information about a new movie does not influence the time that the individual takes to act on the decision. Since the time to decide is influenced by the individual's media habits and word-of-mouth behavior, while the time to act is influenced by a different set of factors (such as the individual's moviegoing habits, free time, and willingness to travel to a convenient theater to see a new movie), there is no a priori reason to hypothesize a strong dependence between these two subprocesses. However, this is an empirical issue that certainly deserves further attention.[1]

We also follow Berg (1981) by assuming that any individual in a potential adopter population of size N may, independently of other members of the population, be exposed to the influential piece of information in any interval $(t, t + \Delta t)$ with probability $\lambda\Delta t + o(\Delta t)$. Hence, the probability that the time to decide will occur in $[0, t]$ for any member of the population is exponential with a stationary[2] parameter λ, and that the expected time to get exposed to the influential piece of information (and hence the expected time to decide to see a new movie) for any individual is $1/\lambda$. When λ is extremely high, the average time to decide approaches zero. We would expect λ to be particularly large for "blockbuster" movies (such as the Steven Spielberg mega-hit *Jurassic Park*), where the intensity of information in the marketplace is very high.

Once the individual has decided to adopt, s/he is assumed to wait for a certain time before actually adopting (seeing) the movie (i.e., the time to act). We assume that the time to act is exponentially distributed with a stationary parameter γ. We recognize that the stationarity assumption may be somewhat limited in realism, but the stationary time-to-act parameter-based model is appealing from a forecasting perspective because of its parsimony. Such a model also allows us to illustrate some interesting basic properties that are unique to our modeling approach. In a later section of the paper, we relax the stationarity assumption with respect to the time-to-act subprocess, by allowing γ to be an explicit time-varying function of exhibition intensity.

Theoretically, the expected time to act can approach zero as the time-to-act parameter γ approaches infinity. In practice, however, a very intensive exhibition strategy can reduce the expected time to act to a minimum nonzero value, since we would always expect a finite time to act upon the adoption decision for reasons unrelated to exhibition (e.g., consumer's free time availability). Hence, we would expect the parameter γ to be bounded from above by a finite value.

The model notation is as follows.

$X(T)$ = CDF for the event that any individual will *decide* to adopt (see) the new movie by time T

$Y(\tau)$ = CDF for the event that any individual will *act* on the decision to adopt (see) the new movie by time τ, given that the individual has decided to adopt at $\tau = 0$

$t = T + \tau$ = Time to *adopt* (see) the new movie (sum of time to *decide* and time to *act*)

$Z(t)$ = CDF for the event that any individual actually will see (adopt) the new movie by time t

N = Size of population of interest (constant)

$N(t)$ = Cumulative number of adopters by time t

$E[N(t)]$ = Expected number of cumulative adopters by time t

λ = Time-to-decide parameter

γ = Time-to-act parameter

Based on the assumptions specified earlier, the two stochastic processes have the following densities and Cumulative Distribution Functions:

[1] To provide preliminary empirical support for the independence assumption, we collected data from 139 student respondents on their TV as well as print media usage frequency, which we used as indicator variables for the unobservable time-to-decide construct, since, *ceteris paribus*, heavy media users would be exposed earlier to motivating new movie information. We also asked respondents to estimate the average time (in weeks) that they took to act on a decision to see a new movie, *after* they had decided to see the movie, which we used as a direct estimate of the time-to-act construct. The correlation between the TV media usage frequency and the time to act was -0.129 ($p = 0.130$, $N = 139$). The correlation between print media usage frequency and the time to act was -0.091 ($p = 0.284$, $N = 139$). These statistically insignificant correlations indicate that individuals who take less time to decide to see a movie do not necessarily act upon the decision any earlier than individuals who take longer to decide.

[2] To assess empirically the validity of the stationarity assumption for the time-to-decide parameter, we collected actual data on times to decide for four movies from a diary panel of 64 student participants, who were surveyed longitudinally over a 10-week period on what movies they planned to see during each week. The time to decide was defined as the time interval between the week that the individual recorded his decision to see a particular movie, and the *first* week that the advertising became available for that movie (TV and print advertising for these movies was also monitored over the 10-week period). Each observation thus represents the *week* during which the individual decided to see a particular movie. The panel data yielded a total of 151 observations of the time to decide, pooled across respondents, and across movies. An exponential functional form fitted to the observed histogram of times-to-decide yielded an R^2 of 0.988. A goodness-of-fit test to determine whether the observed probability distribution of times-to-decide is consistent with an exponential probability distribution function with a stationary rate parameter yielded a test statistic (χ^2 with 5 d.f.) of 6.01, which is less than $\chi^2_{5,0.05} = 11.05$. Hence, we find empirical evidence suggesting that the time-to-decide random variate can be approximated by an exponential distribution with a stationary parameter.

$$x(T) = \lambda e^{-\lambda T}; \quad X(T) = 1 - e^{-\lambda T}, \tag{1}$$

$$y(\tau) = \gamma e^{-\gamma\tau}; \quad Y(\tau) = 1 - e^{-\gamma\tau}. \tag{2}$$

Since the time to adopt is the sum of the time to decide and the time to act ($t = T + \tau$), the cumulative distribution function for the random variable t denoting the time to adopt is the convolution of the CDFs of the time to decide and the time to act (Parzen 1962, p. 16). That is,

$$Z(\cdot) = X(\cdot) * Y(\cdot). \tag{3}$$

The CDF for $Z(\cdot)$ is therefore given by

$$Z(t) = \int_0^t X(u)dY(t-u). \tag{4}$$

Substituting for $X(\cdot)$ and $dY(\cdot)$ from (1) and (2) respectively, (4) can be written as

$$Z(t) = \int_0^t (1 - e^{-\lambda u})[\gamma e^{-\gamma(t-u)}]du. \tag{5}$$

The final expression for the CDF for the time to adopt is given by (see Sawhney and Eliashberg 1994 for the derivation)

$$Z(t) = \frac{1}{\lambda - \gamma}[(\lambda - \gamma) + \gamma e^{-\lambda t} - \lambda e^{-\gamma t}]. \tag{6}$$

The above CDF can be recognized as the CDF of the Generalized Gamma distribution, which is a generalization of the Gamma distribution that arises when two independent exponential random variates with differing rate parameters are convolved (McGill and Gibbon 1965).

So far, we have derived the dynamic adoption probability for an individual consumer. However, we are ultimately interested in calculating the expression for the cumulative number of adopters as a function of time. Since the unordered points of time at which each of the individual consumers makes the decision to adopt are randomly and identically distributed, the distribution of the cumulative number of adopters by time, $N(t)$, can be written as

$$N(t) \sim \text{Binomial}(N, p), \quad \text{where } p = Z(t). \tag{7}$$

The expected value of the cumulative number of adopters is then the expected value of the above Binomial distribution:

$$E[N(t)] = N \cdot Z(t). \tag{8}$$

Substituting for $Z(t)$ from Equation (6), the expression for the expected number of cumulative adopters at time t becomes

$$E[N(t)] = \frac{N}{\lambda - \gamma}[(\lambda - \gamma) + \gamma e^{-\lambda t} - \lambda e^{-\gamma t}]. \tag{9}$$

The expression for the rate of expected adoption can be derived by differentiating the above expression with respect to time:

$$\frac{\partial}{\partial t} E[N(t)] = \frac{N\lambda\gamma}{\lambda - \gamma}[e^{-\gamma t} - e^{-\lambda t}]. \tag{10}$$

Further, since the cumulative adoption is given by a Binomial distribution with the parameters in Equation (7), the variance of the cumulative adoption is simply given as

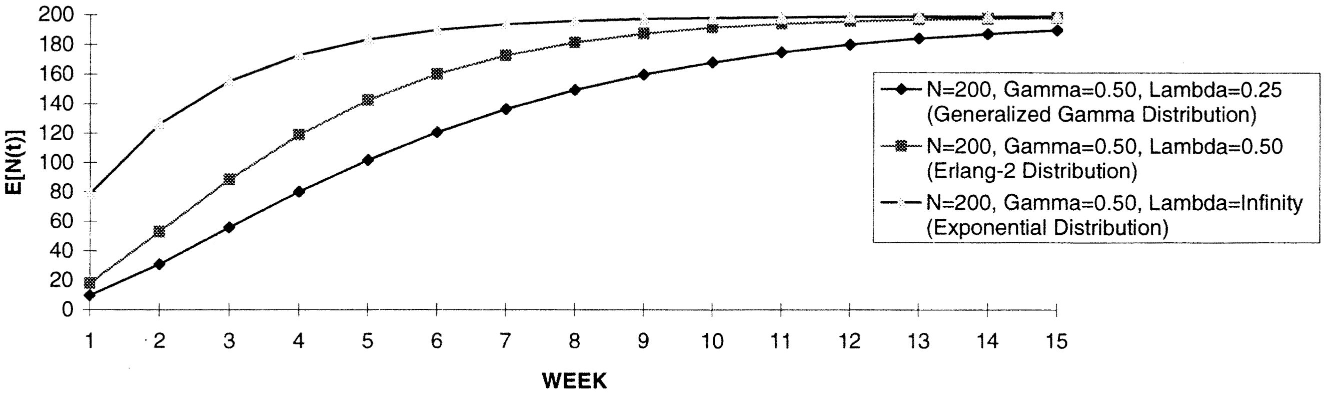
$$\text{Var}[N(t)] = N \cdot p(t) \cdot [1 - p(t)] = N \cdot Z(t) \cdot [1 - Z(t)]. \tag{11}$$

Substituting for $Z(t)$ from (6) and simplifying, the variance expression can be written as

$$\text{Var}[N(t)] = \frac{N}{(\lambda - \gamma)^2}[(\lambda - \gamma) + \gamma e^{-\lambda t} - \lambda e^{-\gamma t}] \times [\lambda e^{-\gamma t} - \gamma e^{-\lambda t}]. \tag{12}$$

Simulated patterns for the expected cumulative adoption (Equation 9), the rate of expected adoption (Equation 10), and the Coefficient of Variation ($\text{Var}[N(t)]^{1/2}/E[N(t)]$) for the stationary parameters model are illustrated in Figures 1(a), 1(b), and 1(c) respectively for different values of the time-to-decide parameter λ ($\lambda = 0.25$, 0.50, and ∞), holding the values of γ and N constant ($\gamma = 0.50$, $N = 200$). The relative values of γ and λ determine the shape of the adoption pattern. In particular, when λ is finite and $\lambda \neq \gamma$, the rate of expected adoption may be nonmonotonic, and the p.d.f. of the time to adopt is best represented by the three-parameter Generalized Gamma distribution (Equation 6). When λ is finite and $\lambda = \gamma$, the rate of expected adoption may be nonmonotonic, and the p.d.f. of the time to adopt is adequately represented by a two-parameter Erlang-2 distribution (with $z(t) = \gamma^2 te^{-\gamma t}$). Finally, when $\lambda \to \infty$, the rate of expected adoption is

Figure 1(a) **Simulated Patterns for Stochastic Adoption Model: Expected Cumulative Adoption**



monotonically decreasing, and the p.d.f. of the time to adopt is adequately represented by a two-parameter exponential distribution (with $z(t) = \gamma e^{-\gamma t}$). In summary, the stationary parameters model can accommodate a wide range of adoption patterns. And, as Figure 1(c) illustrates, the model also provides estimates of the levels of uncertainty associated with the different expected adoption patterns, measured in terms of the coefficient of variation.

2.2. Model Properties

2.2.1. Effect of Time-to-Decide Parameter on Rate of Expected Adoption. The equation for the rate of expected adoption (10) is the difference of two exponential terms. The exponential term with the parameter λ corresponds to the time-to-decide, while the exponential term with the parameter γ corresponds to the time-to-act sub-process. The time-to-decide sub-process "dampens" the adoption pattern, which would become a pure exponential decay pattern ($N\gamma e^{-\gamma t}$), whenever the time-to-decide parameter (λ) becomes extremely large. This is conceptually similar to the Fourt and Woodlock (1960) model. Conversely, as $\lambda \to 0$ (corresponding to infinite average time-to-decide), the rate of the expected adoption approaches zero. For finite values of λ, as the parameter decreases (i.e., as average time-to-decide $1/\lambda$ increases), the expected cumulative adoption and the expected adoption rate become relatively slow.

2.2.2. Effect of Model Parameters on Uncertainty in Expected Adoption. From Figure 1(c), we observe qualitatively that the uncertainty surrounding the expected cumulative adoption pattern, as measured by the Coefficient of Variation ($[\mathrm{Var}\, N(t)]^{1/2}/E[N(t)]$), is higher in the early stages of the life cycle. This is partially due to the fact that the expected cumulative adoption $E[N(t)]$ (the denomi-

Figure 1(b) **Simulated Patterns for Stochastic Adoption Model: Rate of Expected Adoption**

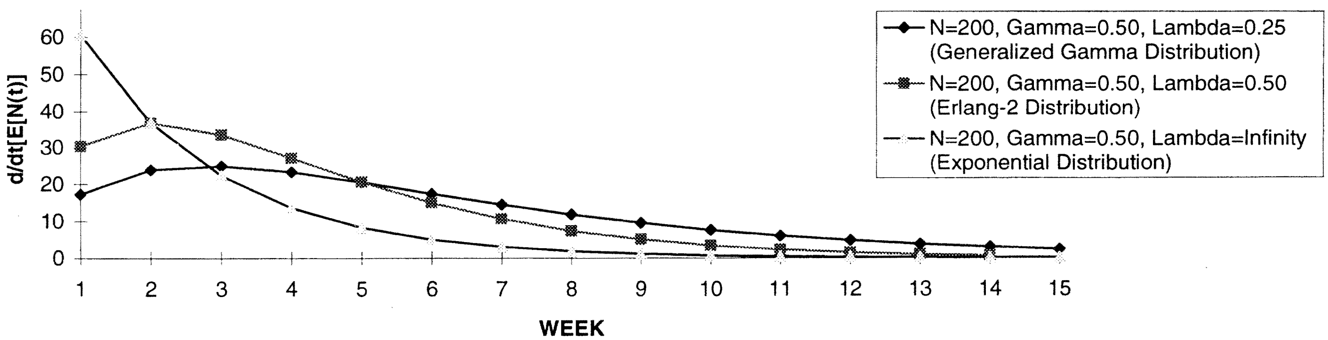
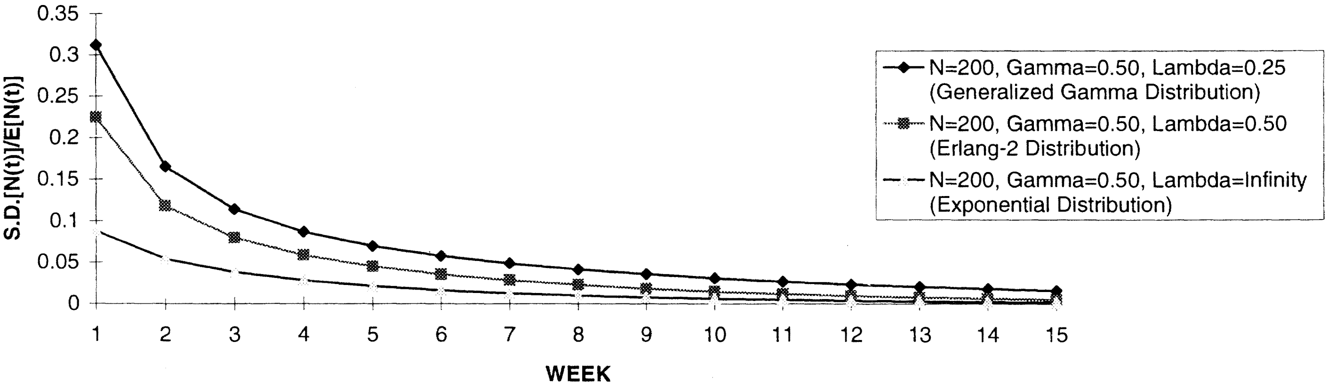


Figure 1(c) Simulated Patterns for Stochastic Adoption Model: Coefficient of Variation



nator term) is relatively small in the early stages and reduces as we go further out into time. This property of the model is consistent with conventional wisdom in the motion picture industry, which states that the performance of a movie in the first few weeks is crucial to determining the eventual success or failure of the movie, but the uncertainty is much higher in the first few weeks. Further, the relative uncertainty is higher for the three-parameter Generalized Gamma model ($\lambda = 0.25$, $\gamma = 0.5$ in Figure 1(c)), compared with the two-parameter Erlang-2 and exponential models ($\lambda = 0.5$, $\gamma = 0.5$, and $\lambda \to \infty$, $\gamma = 0.5$, respectively in Figure 1(c)).

2.3. Model Extensions: Controlling the Time-to-act Parameter

So far, we have assumed that the time-to-act parameter γ is stationary and constant. An important extension of the model would be to relax the stationarity assumption by allowing this parameter to depend on marketing policies, in a fashion akin to incorporating advertising and price into the Bass (1969) diffusion model parameters (see, for example, Kalish and Sen 1986). In our case, the exhibition intensity may change on a week-to-week basis, manifested as the change in the number of screens at which a new movie is shown. The extant exhibition is likely to influence the consumers' time-to-act. In this section, we briefly sketch out an extension of the model that accommodates this phenomenon and discuss some of its implications.

To incorporate the exhibition intensity into the modeling framework, we need to specify the relationship between the time-to-act parameter, γ, and the exhibition intensity (number of screens). One way of doing this is by making the time-to-act parameter proportional to the exhibition intensity. That is,

$$\gamma(t) = cs(t), \tag{13}$$

where $\gamma(t)$ denotes the nonstationary exhibition-related time-to-act parameter at time t, $s(t)$ denotes the nonstationary exhibition intensity at time t, and c denotes a proportionality factor.

The exhibition intensity $s(t)$ can be operationalized either as the *absolute* number of screens that the movie is showing at, or the *share* of total screens for the new movie. Distributors in the motion picture industry tend to follow one of two generic exhibition strategies, called the "wide release" (or blitz) strategy and the "platform release" strategy. The former exhibition strategy is generally used by major studios for movies featuring well-recognized stars that are intensively promoted. The wide release strategy is characterized by a high level of initial exhibition intensity, which gradually decays over time as demand becomes saturated. The platform release strategy, on the other hand, is characterized by a low initial exhibition intensity, which gradually increases over time (capitalizing upon positive word-of-mouth) before eventually declining as demand is saturated. The platform release strategy generally is employed by smaller independent distributors, although it is gaining popularity with major studios for "sleeper" movies with relatively complex storylines (recent examples include *Leaving Las Vegas*, *Sense and Sensibility*, and *Unstrung Heroes*). To capture the wide release pattern (decreasing exhibition intensity), one can invoke the following time-dependent functional form[3] for the exhibition intensity $s(t)$:

$$s(t) = s_0 e^{-bt}, \tag{14}$$

where s_0 denotes the number of opening screens for the movie, and b denotes a decay parameter that determines the rate of decline of the exhibition intensity. An examination of movie exhibition intensity patterns indicates that the exponential form adequately represents a majority of empirically observed exhibition patterns for movies following the wide release strategy (Sawhney and Eliashberg 1994).

Using (13) and (14), the expression for the time-varying distribution-related relay parameter $\gamma(t)$ can be written as $\gamma(t) = (cs_0)e^{-bt} = \gamma e^{-bt}$, (where $\gamma = cs_0$).

The final expression for the expected number of cumulative adopters under the wide release pattern (Equation 14) becomes (see Sawhney and Eliashberg 1994):

$$E[N(t)] = N\left\{1 + e^{-\lambda t} \sum_{i=1}^{\infty} \frac{\gamma^i}{\prod_{k=1}^{i} (\lambda - kb)} - \exp\left[-\frac{\gamma}{b}(1 - e^{-bt})\right] \times \left[1 + \sum_{i=1}^{\infty} \frac{(\gamma e^{-bt})^i}{\prod_{k=1}^{i} (\lambda - kb)}\right]\right\}. \tag{15}$$

The expression for the rate of expected adoption for the wide release pattern can be evaluated by differentiating the above expression with respect to time:

$$\frac{\partial}{\partial t} E[N(t)] = N\lambda\left\{-e^{-\lambda t} \sum_{i=1}^{\infty} \frac{\gamma^i}{\prod_{k=1}^{i} (\lambda - kb)} + \exp\left[-\frac{\gamma}{b}(1 - e^{-bt})\right] \times \left[\sum_{i=1}^{\infty} \frac{(\gamma e^{-bt})^i}{\prod_{k=1}^{i} (\lambda - kb)}\right]\right\}. \tag{16}$$

The adoption patterns for the wide release pattern (decreasing distribution intensity) suggest the following properties:

(a) Stationary Time-to-act Parameter Model as a Special Case. The stationary time-to-act parameter model is a special case of the decreasing intensity model, as the exhibition decay parameter $b \to 0$, so that the nonstationary distribution-related time-to-act parameter $\gamma(t)$ reduces to a constant value γ (where $\gamma = cs_0$).

(b) Effect of Decay Parameter b on Adoption Pattern. A faster decay in exhibition intensity (corresponding to larger values of b) results in a slower expected rate of adoption. This is intuitive, since the limited availability of the new movie slows down the adoption process.

(c) Coefficient of Variation. A faster decay in exhibition intensity is associated with a higher coefficient of variation (C.V.). Further, unlike the stationary model where the C.V. approaches zero in the steady state, the C.V. *never goes to zero* in the steady state for the decreasing intensity (wide distribution strategy) model. This is due to the fact that the expected cumulative adoption probability does not approach one in the steady state for the decreasing intensity model, a property discussed below.

(d) Steady-state Value of Expected Cumulative Adoption. In the stationary parameter case, the steady-state value of expected adoption is N. This implies that the entire potential adopter population eventually sees the new movie. However, in the case of decreasing exhibition intensity, it seems intuitive that the upper bound of the cumulative adoption should be lower than N, since some individuals may have decided to see the movie but it takes them too long to act on the decision, and hence they may not be able to see the movie, since the movie may no longer be playing at nearby theaters. From Equation (15), as $t \to \infty$, the expression for the upper bound of cumulative adoption (the steady-state value of $E[N(t)]$) reduces to $N[1 - e^{-\gamma/b}]$, which is always less than N.

To summarize, the modeling framework presented in this paper can accommodate various aspects related to moviegoers' decision to watch a movie, their implementation of the decision, and the extent to which one can control the exhibition intensity of the movie, which in turn will impact the consumer's time-to-act parameter. A family of models that involve successive refinements of the framework's basic components is discussed in an unabridged version of this

[3] This functional form does not address the platform release strategy. The platform release strategy can be modeled using an exponential functional form such as $s(t) = \alpha[d - e^{-ct}]$ (Sawhney and Eliashberg 1994).

paper (Sawhney and Eliashberg 1994). These models are more elaborate and realistic in terms of the assumptions underlying the functional specifications. However, the richness comes at the expense of reduced parsimony and the inability to forecast early the demand pattern for movies.

3. Empirical Testing

In this section, we present forecasting results for the most parsimonious version of the family of forecasting models that can be generated by our modeling approach—the model with stationary time-to-decide and time-to-act parameters (henceforth called BOXMOD-I). We collected data on the weekly box-office performance of a sample movies from *Variety*, a leading trade publication in the entertainment industry. We took care to include commercially successful as well as unsuccessful movies. These movies were released during 1990 and 1991. Since movie admission ticket prices are relatively uniform, we conduct analyses in terms of gross box-office dollar revenues rather than in terms of attendance. The attendance figures can be calculated by simply dividing the revenues by the average admission price (which is approximately $5.00). In presenting the forecasting results, we compare the forecasting performance of BOXMOD-I to other benchmark or null models.

3.1. Early (Three-week-based) Forecasting

3.1.1. Estimation Details. We estimate BOXMOD-I using a standard nonlinear least squares procedure. In particular, we use the NONLIN procedure in SYSTAT 5 for Windows (Systat 1992). The procedure uses a multidimensional simplex algorithm to minimize a loss function. The loss function is the sum of the squared deviations between the predicted and actual grosses for each of the weeks in the calibration period. We constrain the loss function to ensure nonnegativity and finiteness of the three model parameters N, γ, and λ. In every case, the procedure converges quickly, and the parameter estimates are robust with respect to the starting values for the parameter estimates, indicating that the procedure produces global minima for the loss function.

3.1.2. Benchmark Models. We compared the forecasting performance of BOXMOD-I with two benchmark models. The first benchmark model we use is the Bass (1969) model.[4] Like BOXMOD-I, the Bass model has three parameters and hence can be calibrated with a minimum of three weeks of data. To maintain comparability of estimation procedures, we estimated the Bass model using the nonlinear estimation procedure suggested by Srinivasan and Mason (1986).

The second benchmark model we use is a "Nonlinear Screens Model" (henceforth called the NSM), which assumes that box-office revenues are determined primarily by the intensity of distribution. The NSM, unlike the time-based Bass (1969) model and the time-based BOXMOD-I, is formulated in terms of a (nonlinear) distribution intensity response curve. It thus represents a conservative and strong test for the latter two models since it incorporates information that is absent in the other two models. We use a logistic functional form for the response curve. The logistic functional form is logically consistent and intuitively appealing. With appropriate scaling of parameters, the logistic functional form produces zero revenues at zero distribution intensity. In addition, revenues saturate at a finite level at infinite distribution intensity, and there are diminishing marginal returns to increasing distribution intensity. Further, the logistical functional form assures nonnegativity of revenues. The NSM that we use as a benchmark model is specified as follows.

$$R(s_j) = \frac{e^{\alpha_j s_j} - 1}{(1 + \beta_j)(1 + \beta_j e^{\alpha_j s_j})}, \tag{17}$$

where s_j denotes the distribution intensity for movie j (measured as the weekly number of screens at which the movie is showing), and $R(s_j)$ denotes weekly B.O. revenues for movie j produced at distribution intensity level s_j.

[4] It could be argued that a stronger test would be to employ the Bass model with covariate effects, where distribution intensity would shift the baseline hazard function up or down, following the Generalized Bass Model (GBM) formulation suggested by Bass et al. (1994). This would not constitute a fair test since BOXMOD-I does not incorporate covariates. Moreover, since empirical distribution time paths for most movies tend to be exponential, the plain Bass model will produce similar results to the GBM, since the percentage change in distribution intensity from period-to-period is almost constant (see Bass et al. 1994 for details).

To make a fair comparison of forecasting performance, we calibrated the NSM with only three weeks of screens data, just like BOXMOD-I and the Bass model. The NSM's two parameters (α and β) were estimated using the same NONLIN procedure as for BOXMOD-I. The remaining ($T_j - 3$) weeks were used as the holdout period, where T_j is the length of movie j's life cycle in weeks.

3.1.3. Forecasting Results—Expected Cumulative Revenues. Forecasting results for a sample of 19 movies are reported in Table 1 for BOXMOD-I and the two benchmark models. The rate of expected adoption for two representative movies (*Terminator*-2 and *Betsy's Wedding*) are illustrated in Figure 2(a) and Figure 2(b) respectively. In Table 1, the third column lists the actual cumulative Box-Office revenues for each movie over the entire time period of T_j weeks. We next report the parameter estimates and two performance indices—the Mean Squared Error (MSE) and the absolute percentage error (Abs. Error) in the predicted cumulative box-office revenues for BOXMOD-I and the two benchmark models.

BOXMOD-I performs very well in absolute as well as in relative terms with a three-week calibration period, and outperforms both benchmark models. The absolute percentage predictive error is within 10% for eight out of 19 cases for BOXMOD-I, as against five out of 19 cases for the Bass model, and three out of 19 cases for the NSM. The average predictive error across the sample is 11.23% for BOXMOD-I, while it is 16.06% for the Bass model, and 22.76% for the NSM. The average MSE for BOXMOD-I is 1.659, which compares favorably with an average MSE of 2.325 for the Bass model, and 12.774 for the NSM. Further, in the three cases where BOXMOD-I does not provide valid parameter estimates with three data points, neither does the Bass model, which suggests that the problem lies with the data, and not with the model. Upon closer examination of these cases, we indeed found that two of these movies had an aberrant data point in the first three weeks. Finally, we note that in the cases where the Bass model estimates suggests a two-parameter exponential model (i.e., movies where $q = 0$), BOXMOD-I also suggests a two-parameter exponential model (with $\lambda \to \infty$) with an identical rate parameter. Our interpretation of these cases is that these are intensively promoted, discussed, and advertised movies. This is consistent with the interpretation of the dominant external influence Fourt and Woodlock (1960) model.

An explanation for the relatively poor forecasting performance of the NSM is in order here. Since exhibition contracts typically lock in exhibitors for a minimum run length of three to four weeks, there is little variability in the distribution intensity for most motion pictures in the first three weeks, which results in unstable parameter estimates for the NSM with the first three weeks as the calibration period. While the NSM is a very good model in terms of the goodness-of-fit to actual distribution response curves (the average absolute percentage error for the same sample of movies is 5.31% using all T_j weeks to calibrate the NSM), the NSM has limited utility for early B.O. forecasting. For early forecasting, BOXMOD-I outperforms both the Bass model and the NSM.

Examining the BOXMOD-I's parameter values across movies, we find in general that big-budget movies that were intensively promoted prior to release (e.g., *Terminator-2, Robin Hood, Naked Gun* $2\frac{1}{2}$) tend to show exponentially declining B.O. revenues patterns with very large time-to-decide parameters λ (and hence very short average time-to-decide by moviegoers). On the other hand, movies that were less intensively promoted prior to launch and relied on positive word-of-mouth (e.g., *Ghost*, *Betsy's Wedding*, and *Arachnophobia*) tend to exhibit nonmonotonic B.O. revenues patterns best represented by the two-parameter Erlang-2 distribution or the three-parameter Generalized Gamma distribution, with smaller time-to-decide parameters (and hence longer average time-to-decide by moviegoers). Movies that maintained exhibition ("legs" in industry terminology) for a long time period (T_j) and were successful at the box-office (e.g., *Robin Hood*, *Ghost*, *Doc Hollywood*) tend to have lower values of the time-to-act parameter γ.

3.1.4. Relative Uncertainty in Expected Revenues. Since we use a stochastic modeling framework, in addition to forecasting expected cumulative revenues, BOXMOD-I can produce quantitative estimates of the *uncertainty* surrounding the expected revenue forecasts. In order to compare the uncertainty in expected revenue forecasts across a set of movies on a comparable basis, we measure the relative uncertainty in terms of the dynamic value of the Coefficient of Variation (C.V.) for the

Table 1 Comparison of Forecasting Performance of BOXMOD-I Against Benchmark Models

			BOXMOD-I[5]							Bass Model						Nonlinear Screens Model				
Movie Name	T_i[1] (Wks.)	Actual Cum. B.O. ($m)	N	γ	λ	Type of Pattern	Pred. Cum. B.O. ($m)	MSE[3]	Absolute[4] Error (%)	p	q	m	Pred. Cum. B.O. ($m)	MSE	Absolute Error (%)	α	β	Pred. Cum. B.O. ($m)	MSE	Absolute Error (%)
Terminator 2	24	199.858	187.875	0.553	42.328	Exponential	142.536	11.456	28.68%	0.553	0.000	142.532	142.532	12.726	28.68%	1.415	0.005	294.701	58.417	47.46%
Robin Hood	20	164.384	166.126	0.319	57.488	Exponential	141.582	1.845	13.87%	0.319	0.000	141.780	141.540	1.845	13.90%	1.405	0.017	213.607	11.660	29.94%
The Rocketeer	17	46.655	49.377	0.485	1.670	Gen. Gamma	46.230	0.107	0.91%	0.347	0.371	42.804	42.804	0.094	8.26%	****	****	****	****	****
Dying Young	10	32.372	42.880	0.560	77.672	Exponential	32.102	0.057	0.84%	0.560	0.000	32.218	32.099	4.936	0.84%	0.544	−0.336	34.583	3.991	6.83%
Naked Gun $2\frac{1}{2}$	19	86.928	97.910	0.557	77.438	Exponential	73.689	1.197	15.23%	0.557	0.000	73.703	73.701	4.180	15.22%	0.392	−0.353	91.832	8.848	5.64%
The Doctor	21	37.887	****[2]	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****
V.I. Warshawski	10	11.127	11.460	1.310	1.310	Erlang-2	9.953	0.031	10.55%	0.553	0.858	9.607	9.607	0.246	13.66%	0.680	−0.280	12.198	0.291	9.63%
Mobsters	10	19.760	21.259	0.739	2.985	Gen. Gamma	17.961	0.062	9.10%	0.651	0.161	17.801	17.794	0.472	9.95%	0.562	−0.315	21.918	1.337	10.92%
Hot Shots!	16	68.258	84.043	0.279	29.404	Exponential	72.714	0.209	6.53%	0.279	0.000	73.562	72.713	0.121	6.53%	****	****	****	****	****
Doc Hollywood	19	54.617	72.244	0.193	33.644	Exponential	64.191	0.476	17.53%	0.193	0.000	65.883	64.199	0.238	17.54%	****	****	****	****	****
Die Hard 2	15	114.218	118.801	0.427	2.908	Gen. Gamma	108.009	0.327	5.44%	0.398	0.149	102.719	102.680	1.031	10.10%	****	****	****	****	****
Days of Thunder	13	81.443	84.149	0.435	1.439	Gen. Gamma	79.624	0.071	2.23%	0.295	0.421	71.384	71.368	1.139	12.37%	****	****	****	****	****
Betsy's Wedding	10	19.674	24.933	0.585	0.585	Erlang-2	23.855	0.340	21.25%	0.199	0.724	18.949	18.940	0.011	3.73%	****	****	****	****	****
Exorcist III	6	24.523	27.572	0.967	0.967	Erlang-2	25.157	0.535	2.59%	0.288	1.353	22.062	22.055	0.390	10.06%	0.329	−0.535	27.914	4.262	13.83%
Arachnophobia	10	50.193	61.985	0.553	0.553	Erlang-2	59.172	1.483	17.89%	0.181	0.876	42.911	42.905	0.955	14.52%	3.319	0.078	79.220	13.390	57.83%
Ghost	20	196.104	154.919	0.347	0.347	Erlang-2	152.355	8.288	22.31%	0.116	1.020	68.601	68.601	8.707	65.02%	****	****	****	****	****
Bird on a Wire	19	70.292	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****
Cadillac Man	12	27.256	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****	****
Wild at Heart	11	14.304	15.558	0.622	0.622	Erlang-2	14.969	0.067	4.65%	0.174	1.346	10.498	10.498	0.117	26.61%	****	****	****	****	****
Average Predictive Errors								1.659	11.23%					2.325	16.06%				12.774	22.76%

[1] T_i is the total length of the life cycle of the movie in weeks. So, if the calibration period is three weeks, the holdout period is $(T_i - 3)$ weeks.

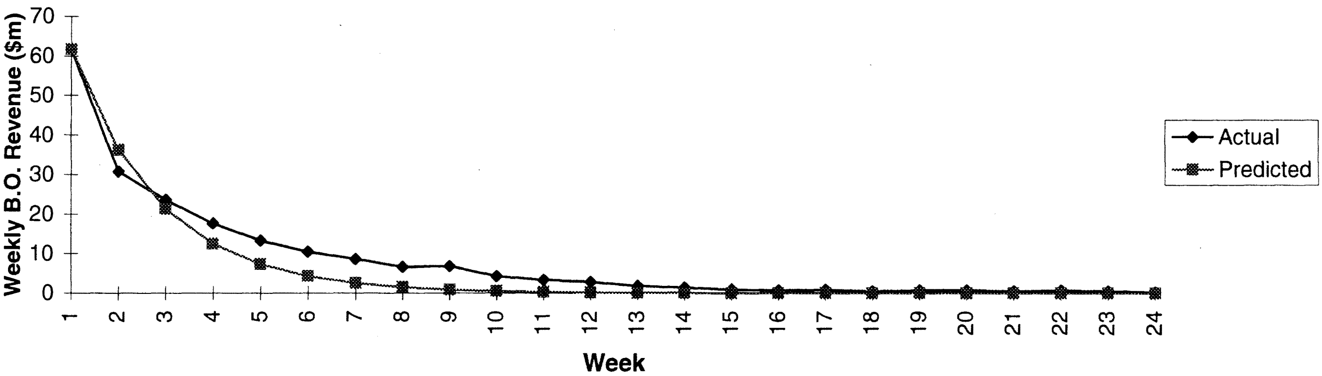
[2] Indicates that the model did not provide finite nonnegative values for one or more model parameters.

[3] Mean Squared Error is calculated for all T_i weeks, including the calibration period of three weeks.

[4] Absolute Error is the absolute value of (Predicted Cum. B.O. − Actual Cum. B.O.)/Actual Cum. B.O., expressed as a percentage.

[5] All models, including BOXMOD-I, Bass model, and the Nonlinear Screens Model (NSM), are calibrated using three weeks of data and a nonlinear estimation procedure.

Figure 2(a) **BOXMOD-I Early Forecasting Results: Weekly Box-Office Revenues (*Terminator-2*)**



revenue forecasts for each movie. The C.V. is given as $(\mathrm{Var}[N(t)]^{1/2}/E[N(t)])$, where $\mathrm{Var}[N(t)]$ is given by Equation 12, and $E[N(t)]$ is given by Equation 9.

The relative uncertainty in expected cumulative box-office revenues for some movies is illustrated in Figure 2(c). We illustrate the C.V. for two movies (*Terminator-2* and *Die Hard* 2) that followed the wide distribution strategy and exhibited monotonically decreasing B.O. revenue patterns. We also illustrate the C.V. for two movies that exhibited nonmonotonic B.O. revenue patterns and had relatively smaller time-to-decide parameters (*Betsy's Wedding* and *Ghost*). We note that the relative uncertainty is higher for the latter two movies over the entire life cycle, a result that conforms with the industry intuition that, in general, the commercial performance of movies that are released based on a platforming strategy and build exhibition intensity over time is more uncertain than the performance of movies that are widely promoted and intensively distributed.

3.2. Forecasting with No Revenues Data or Little Revenues Data—An Adaptive Approach

We have demonstrated the utility of BOXMOD-I for early forecasting with three weeks of data. However, the model's utility could be greatly enhanced if it could provide initial forecasts with *no box-office revenue data*, which could then be adaptively updated as box-office revenue data gradually become available. One approach to generating sales forecasts with no revenue data is to conduct a meta-analysis of parameter estimates for related products (see, for example, Sultan et al. 1989). In a meta-analysis procedure, the model parameters for several existing products (whose sales histories are available) are systemati-

Figure 2(b) **BOXMOD-I Early Forecasting Results: Weekly Box-Office Revenues (*Betsy's Wedding*)**

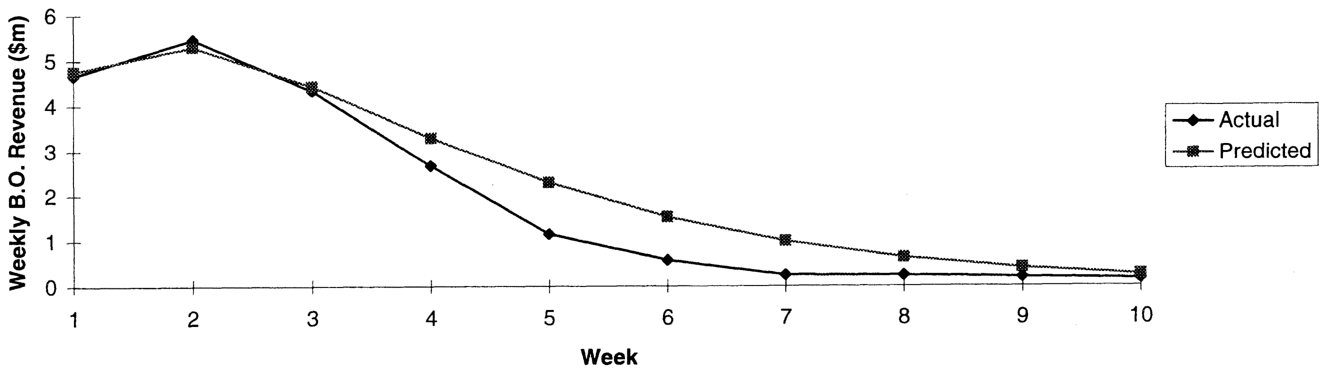
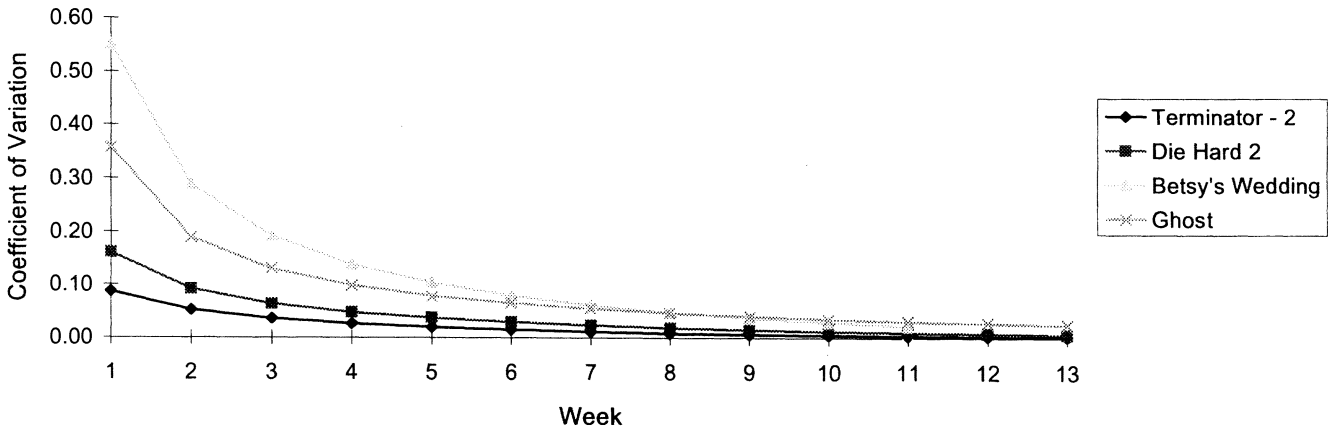


Figure 2(c) **Relative Uncertainty in Expected Cumulative Box-Office Revenues**



cally related to the attributes of the products, and this relationship is used to predict the parameter estimates (and consequently the sales) for any new product. The motion picture industry offers a unique opportunity to conduct such a cross-sectional meta-analysis, since the sales histories for hundreds of movies are publicly available.

3.2.1. Estimation Details. It is in the above noted spirit that we employed an adaptive scheme which draws upon meta-analysis to generate initial estimates for the model parameters (N, λ, γ) before any box-office data are available. However, we extend the traditional meta-analysis procedure in that we update successively the initial estimates by using information contained in early box-office data. We pool the estimates derived from the meta-analysis approach (which uses no box-office data) with estimates derived from box-office data using an adaptive weighing scheme, where the weights reflect the relative precision of these two sets of estimates. The adaptive procedure proceeds as follows:

Step 1—*No Data Available* ($t = 0$). To obtain the *meta-analysis-based initial estimates* for any hold-out movie k ($k \in K$), we first regress the "true" parameters (i.e., parameter values for N, γ, and λ obtained via the NONLIN procedure using all available data for each movie j in a large calibration sample J ($j \in J$), where T_j is the length of its life cycle) against Z, a vector of movie characteristics. Then, we substitute the characteristics of the new movie k into this regression equation to obtain its estimated three parameters $\hat{N}_{0k}$, $\hat{\gamma}_{0k}$, and $\hat{\lambda}_{0k}$. We also estimate the expected error in the estimates for any new movie as the average error of the meta-analysis-based parameter estimates across all movies in the calibration sample. The expected value and the expected error in the initial meta-analysis-based parameter estimates for any new movie k in the holdout sample are given mathematically by

$$\hat{N}_{0k} = \alpha_{N0} + \sum_{i=1}^{I} \beta_{N0i} Z_{ik}; \quad \hat{\sigma}_{N0}^2 = \frac{1}{J} \sum_{j=1}^{J} [N_{T_j} - \hat{N}_{0j}]^2, \tag{18}$$

$$\hat{\gamma}_{0k} = \alpha_{\gamma 0} + \sum_{i=1}^{I} \beta_{\gamma 0i} Z_{ik}; \quad \hat{\sigma}_{\gamma 0}^2 = \frac{1}{J} \sum_{j=1}^{J} [\gamma_{T_j} - \hat{\gamma}_{0j}]^2, \tag{19}$$

$$\hat{\lambda}_{0k} = \alpha_{\lambda 0} + \sum_{i=1}^{I} \beta_{\lambda 0i} Z_{ik}; \quad \hat{\sigma}_{\lambda 0}^2 = \frac{1}{J} \sum_{j=1}^{J} [\lambda_{T_j} - \hat{\lambda}_{0j}]^2. \tag{20}$$

In the above equations, $\hat{\sigma}_{N0}^2$, $\hat{\sigma}_{\gamma 0}^2$, and $\hat{\sigma}_{\lambda 0}^2$ represent the expected errors in the initial meta-analysis-based estimates for the parameters N, γ, and λ, respectively.

Step 2—*One Week of Data Available* ($t = 1$). After one week, one data point becomes available, and we can combine the initial meta-analysis-based estimates with estimates based on the B.O. revenues data. The one-week *data-based estimates* $\hat{N}_{1k}$, $\hat{\gamma}_{1k}$, and $\hat{\lambda}_{1k}$ are calculated by regressing the first week's revenues R_{1j} ($j \in J$) against the "true" model parameters across all movies in the calibration database J, and then substituting the new movie's first week's B.O. revenues in the regression equation. The expected errors in the one-week data-based estimates are calculated in a similar fashion as described in *Step* 1. The expected values and the expected errors in the data-based estimates for any new movie k in the holdout sample are given by

$$\hat{N}_{1k} = \alpha_{N1} + \delta_{N1} R_{1k}; \quad \hat{\sigma}^2_{N1} = \frac{1}{J} \sum_{j=1}^{J} [N_{T_j} - \hat{N}_{1j}]^2, \tag{21}$$

$$\hat{\gamma}_{1k} = \alpha_{\gamma 1} + \delta_{\gamma 1} R_{1k}; \quad \hat{\sigma}^2_{\gamma 1} = \frac{1}{J} \sum_{j=1}^{J} [\gamma_{T_j} - \hat{\gamma}_{1j}]^2, \tag{22}$$

$$\hat{\lambda}_{1k} = \alpha_{\lambda 1} + \delta_{\lambda 1} R_{1k}; \quad \hat{\sigma}^2_{\lambda 1} = \frac{1}{J} \sum_{j=1}^{J} [\lambda_{T_j} - \hat{\lambda}_{1j}]^2. \tag{23}$$

In the above equations, R_{1k} denotes the first-week box-office revenue of each movie of interest k ($k \in K$); and $\hat{\sigma}^2_{N1}$, $\hat{\sigma}^2_{\gamma 1}$, and $\hat{\sigma}^2_{\lambda 1}$ are the estimated errors in the parameter estimates derived from one week of data for N, γ, and λ, respectively.

The initial meta-analysis-based estimates and the one-week data-based estimates can now be pooled by weighing each of the estimated parameters in proportion to the reciprocal of the expected errors of the estimates. This weighing scheme is consistent with the literature on linear combination of forecasts that have been obtained from independent sources (Bates and Granger 1969, Newbold and Granger 1974). Bates and Granger (1969) prove that the variance of errors in the combined forecast is minimized when the two forecasts are assigned weights in inverse proportion to the variance of errors for the individual forecasts. For instance, for the parameter N, the pooled estimate and the weights attached to the two estimates are

$$\tilde{N}_{1k} = w_{N0} \hat{N}_{0k} + (1 - w_{N0}) \hat{N}_{1k};$$
$$\text{where } w_{N0} = \frac{1/\hat{\sigma}^2_{N0}}{1/\hat{\sigma}^2_{N0} + 1/\hat{\sigma}^2_{N1}}. \tag{24}$$

The pooling equations for the other two parameters are similar.

Step 3—Two Weeks of Data Available. When two weeks of data are available, in some cases it is possible to obtain estimates for the parameters directly from the model. These are cases where the revenue pattern is well approximated by a strict exponential functional form, and a reduced two-parameter model (with parameters N and γ) is adequate to represent the pattern. In these cases, we replace the data-based estimates with *completely data-driven* or *model-based estimates*. In other cases, where a two-parameter model does not provide valid representation, we resort to obtaining data-based estimates by regressing the first *two* weeks of B.O. revenues R_{1k} and R_{2k} against the "true" parameter estimates, and pool these estimates with the initial meta-analysis-based estimates as outlined in Step 2.

When three or more weeks of data are available, all the parameter estimates are obtained directly from the model, using the box-office data, as discussed in §3.1.

3.2.2. Implementation. To implement the adaptive procedure described above, we created a comprehensive database of B.O. revenues for 111 movies released theatrically in 1992. This database was split into a calibration database (J = 101 movies) used to estimate the regression equations for the three parameters of BOXMOD-I, and a holdout database (K = 10 movies) used to test the forecasting performance of the adaptive procedure.

Next, we created a database of movie characteristics for all 111 movies. The selection of movie attributes was guided in part by attributes employed in prior empirical research on predicting box-office performance based on movie attributes (Litman 1983). However, since we want to generate forecasts prior to launch, we excluded attributes whose values are not known before the time of the movie's release, such as Academy Award nominations or Academy Awards won by the movie. In addition, we were constrained, as in prior empirical research (see Litman 1983), to not being able to use promotional spending information or production budget information, which are potentially important omitted variables. Based on these considerations, the variables used as independent variables for the meta-analysis were as follows:

Genre Dummies: Separate dummy variables for Action, Comedy, Drama, Horror, Science Fiction, Children/Animated.[5]

Sexual Content: Dummy for presence/absence of sexual content/nudity as noted in reviews.

Special Effects: Dummy for presence/absence of big budget special effects.

Sequel: Dummy for sequel (1 if movie is sequel, 0 otherwise)

[5] All genre classifications are based on the Microsoft *Cinemania* (1994) CD-ROM.

Table 2 Meta-Analysis Regressions of BOXMOD-I Parameters on Movie Attributes

Predictor Variable (Movie Attributes)	Dep. Variable = N		Dep. Variable = γ		Dep. Variable = λ	
	Coeff.	P-Value	Coeff.	P-Value	Coeff.	P-Value
Constant	25.343	0.1080	0.522	0.0000	43.971	0.0000
Action					8.858	0.0508
Drama			−0.090	0.0815		
MPAA Rating	−7.810	0.0662	0.072	0.0141	−5.821	0.0307
Presence of Major Stars	52.006	0.0000	−0.128			
Sequel	63.674	0.0000				
Critic Reviews	73.376	0.0008	−0.464	0.0018		
Sexual Content					−9.352	0.0307
# of Observations	101		101		101	
Model R^2	0.419		0.224		0.116	
Adjusted R^2	0.395		0.192		0.089	
F-Ratio for Model	17.320		6.933		4.247	
P-Value	0.0000		0.00006		0.0072	

MPAA Rating: Interval Scale for the MPAA rating (G = 1, PG = 2, PG-13 = 3, R = 4).

Major Stars: Dummy indicating presence of major stars. The classification was based on a list of major stars possessing "marquee value" (*Variety*, July 25, 1994).

Critic Reviews: Standardized average "star" evaluations of three sets of critic ratings (Leonard Maltin, L.A./New York critic opinions published in *Variety*, and reviews from 1994 Baseline Movie Guide).

Given the large set of explanatory variables, we followed previous research (Litman 1983) in reducing the number of independent variables by running three separate step-wise regressions between the values of N, γ, and λ for the calibration sample of 101 movies. The results of these step-wise regressions showing the reduced set of significant variables are summarized in Table 2. We then used Equations (18) through (24) to generate the meta-analysis based estimates, the data-based estimates, the expected error in the estimates, and the adaptive weights attached to the estimates for the calibration sample. The adaptive weights are summarized in Table 3. Next, we used the movie characteristics, the two sets of estimates, and the adaptive weights to estimate the values of N, γ, and λ for the holdout sample of 10 movies. Finally, we used the pooled parameter estimates to generate gross box-office revenue forecasts for each of the movies in the holdout sample with no data, one week of data, two weeks of data, three weeks

Table 3 Adaptive Weights for Meta-Analysis, Data-Based, and Model-Based Estimates

Model Parameter	No Data	One Week		Two Weeks (3 Param. Model)		Two Weeks (2 Param. Model)		Three Weeks (3 Param. Model)
	W_0	W_0	W_1	W_0	W_2	W_0	W_2	W_3
N	1.000	0.308	0.692	0.270	0.730	0.000	1.000	1.000
γ	1.000	0.545	0.455	0.515	0.485	0.000	1.000	1.000
λ	1.000	0.499	0.501	0.498	0.502	NA[1]	NA	1.000

[1] Estimates are obtained from two-parameter model, so there is no estimate for the third parameter λ.

of data, and all T_k weeks of data, where T_k is the length of movie k's life cycle.

3.2.3. Discussion of Results of Adaptive Forecasting Scheme. From Table 2 we observe that the movie attributes are useful predictors of the maximum possible box-office potential N ($R^2 = 0.419, p < 0.00001$). This predictive power compares favorably with Litman (1983), who reports an R^2 of 0.485 in predicting B.O. revenues accruing to distributors from movies released in 1972–1978 using similar movie attributes, especially in view of the fact that, unlike him, we do not include variables related to Academy Awards in our list of predictor variables. The predictive power of the movie attributes for the time-to-act parameter γ is more limited ($R^2 = 0.224, p < 0.0001$), while the movie attributes are only weakly predictive of the time-to-decide parameter λ ($R^2 = 0.116$, $p = 0.0072$). The low predictive performance for the time-to-decide parameter could be partly attributed to the omission of variables such as the advertising spending, which influence the time-to-decide parameter. Advertising spending has been found to be an important determinant of motion picture box-office performance (Sawhney and Eliashberg 1995).

The coefficients for the predictor variables across the regression equations yield interesting insights. From the signs of the coefficients in the regression for N, we note that presence of major stars, awareness inherited by sequels, and positive critic reviews serve to enhance a new movie's ultimate cumulative box-office potential (recall that N measures the box-office potential of the movie *if the movie played indefinitely*). On the other hand, if a new movie has a MPAA rating of "R", its ultimate box-office prospects are likely to be poorer on average. The coefficients in the regression for γ suggest that R-rated movies have higher time-to-act parameters, while movies belonging to the drama genre, movies with major stars, and movies receiving positive critic reviews are associated with lower time-to-act parameters. The latter two relationships may be due to consumers' anticipation that the movies will play for a longer period of time. The coefficients in the regression for λ indicate that action movies tend to influence viewers to decide faster, while R-rated movies and movies with sexual content are associated with smaller parameters and hence with longer average time-to-decide than G or PG-rated movies.

Table 3 summarizes the adaptive weights attached to the meta-analysis-based estimates, the B.O. revenues data-based estimates, and the model-based estimates, as more data become available. We observe that the weights attached to the data-based estimates are far higher for the *scale parameter N*, as compared to the weights attached to the data-based estimates for the *shape parameters* γ and λ. This indicates that early box-office data reveal more about what the ultimate cumulative B.O. potential for a new movie will be, but offer less insight into *how* the box-office revenues will be spread out over the life cycle. This is an interesting finding that would not have been possible to obtain without developing the adaptive scheme.

Table 4 summarizes the forecasts obtained from the adaptive scheme for the 10 movies in the holdout sample. The forecasts are mixed in terms of their accuracy. While the box-office predictions for movies such as *Under Siege* and *Whispers in the Dark* are acceptable, the initial meta-analysis-based estimates (i.e., estimates with no data at all) lead to a significant underprediction of box-office revenues for movies such as *Wayne's World*, *White Men Can't Jump*, *Toys*, and *Twin Peaks*, and significant over-prediction for movies such as *Trespass* and *Wind*.

With the availability of one week of sales data, the average error in the predicted B.O. revenues reduces significantly, but the percentage errors are still quite high in some cases. However, with two weeks of data, there is a dramatic improvement in the accuracy of the forecasts, mainly due to the fact that in nine out of ten cases (except for the movie *Toys*), we were able to use purely model-based estimates. The model-based estimates are more reliable than the B.O. revenues data-based estimates, which is consistent with our expectations, since the model-based estimates use information on the functional form of the B.O. revenue pattern. With three weeks of data, the model predicts box-office grosses quite accurately, with the average predictive error being less than 8%. The striking exception is *Wayne's World*, which picked up additional screens in week 5, which is quite unusual. When all T_k weeks of data are used, the predictive error is less than 2% across the sample of 10 movies, and the *maximum* predictive error is less than 5%. Such goodness-of-fit demonstrates the ability of BOXMOD-I to explain a wide

Table 4 BOXMOD-I Forecasting Results Using Adaptive Weighing Scheme for Parameter Estimation

Movie	T_k (Weeks)	Actual Cum. B.O. ($m)	Estimates Using No Data						Estimates Using 1 Week of Data						Estimates Using 2 Weeks of Data					
			N	γ	λ	Predicted B.O. ($m)	MSE	Abs. % Error	N	γ	λ	Predicted B.O. ($m)	MSE	Abs. % Error	N	γ	λ	Predicted B.O. ($m)	MSE	Abs. % Error
Under Siege	23	83.58	79.74	0.379	29.55	66.42	1.455	20.5%	77.28	0.410	28.63	63.42	46.608	24.1%	89.45	0.285	NA[2]	77.24	6.217	7.6%
Unforgiven	17	74.92	104.47	0.223	20.19	92.17	3.157	23.0%	93.42	0.314	24.05	80.18	0.435	7.0%	92.97	0.338	NA	77.95	0.300	4.0%
White Men Can't Jump	12	71.23	34.87	0.552	20.69	26.80	19.487	62.4%	62.86	0.505	24.20	49.23	5.084	30.9%	72.72	0.381	NA	59.17	1.537	16.9%
Wayne's World	18	117.91	32.00	0.547	26.01	24.56	39.634	79.2%	69.25	0.493	26.94	54.57	18.581	53.7%	83.03	0.384	NA	68.09	11.981	42.3%
Toys	6	21.38	12.10	0.673	26.51	8.55	7.955	60.0%	26.38	0.608	26.80	19.10	1.624	10.7%	34.21	0.561	25.77	25.17	2.410	17.7%
Whispers in the Dark	5	10.32	12.58	0.603	20.19	8.98	0.140	12.9%	19.59	0.578	23.57	14.01	0.776	35.8%	15.03	0.616	NA	10.39	0.008	0.6%
Trespass	7	12.66	30.21	0.491	29.55	23.03	2.712	81.9%	30.01	0.511	28.30	22.76	2.654	79.8%	20.01	0.685	NA	13.83	0.056	9.2%
Wind	7	5.44	27.39	0.486	35.37	20.85	7.633	283.1%	18.59	0.522	31.10	14.03	2.342	157.8%	8.43	0.658	NA	5.91	0.012	8.5%
Twin Peaks:																				
Fire Walk With Me	6	4.16	−1.82	0.694	20.19	****[1]	****	****	****	****	****	****	****	****	6.82	0.753	NA	4.53	0.006	8.8%
Traces of Red	8	3.02	3.46	0.660	20.19	2.51	0.007	16.9%	6.78	0.622	23.47	4.99	0.225	65.0%	3.52	0.623	NA	2.52	0.006	16.6%
Average Errors							9.131	71.1%					8.703	51.6%					2.253	13.2%

Movie	T_k (Weeks)	Actual Cumulative B.O. ($m)	Estimates Using 3 Weeks of Data						Estimates Using All T_k Weeks of Data					
			N	γ	λ	Predicted B.O. ($m)	MSE	Abs. % Error	N	γ	λ	Predicted B.O. ($m)	MSE	Abs. % Error
Under Siege	23	83.58	95.77	0.254	28.96	84.62	4.249	1.2%	94.26	0.261	40.04	82.83	0.174	0.9%
Unforgiven	17	74.92	90.49	0.349	39.41	76.09	0.293	1.6%	91.11	0.346	34.74	76.81	0.290	2.5%
White Men Can't Jump	12	71.23	82.32	0.306	80.60	68.84	0.228	3.4%	84.53	0.290	29.47	71.32	0.173	0.1%
Wayne's World	18	117.91	94.58	0.302	29.53	81.49	6.813	30.9%	134.58	0.176	34.39	118.51	1.075	0.5%
Toys	6	21.38	25.44	0.861	0.86	23.31	0.369	9.0%	24.20	0.921	0.92	22.15	0.285	3.6%
Whispers in the Dark	5	10.32	14.66	0.631	35.73	10.25	0.008	0.7%	13.37	0.630	5.85	10.27	0.008	0.5%
Trespass	7	12.66	19.24	0.752	38.27	13.09	0.023	3.4%	19.32	0.778	67.20	12.86	0.020	1.6%
Wind	7	5.44	8.24	0.680	45.31	5.79	0.009	6.4%	8.11	0.737	45.31	5.54	0.007	1.9%
Twin Peaks:														
Fire Walk With Me	6	4.16	6.68	0.819	96.34	4.32	0.003	3.8%	6.62	0.843	54.32	4.26	0.003	2.3%
Traces of Red	8	3.02	3.62	0.562	99.55	2.68	0.003	11.3%	3.73	0.487	29.55	2.89	0.002	4.5%
Average Errors							1.200	7.2%					0.204	1.8%

[1] Indicates that the method did not yield any positive finite parameter estimates.

[2] Estimates are obtained from two-parameter model, so there is no estimate for the third parameter λ.

variety of motion picture revenue patterns with an acceptable degree of accuracy. The excellent goodness-of-fit also demonstrates that motion picture box-office patterns display a remarkable empirical regularity that can be captured with a parsimonious two- or three-parameters model. This is an interesting empirical generalization. Further, a comparison of parameter estimates obtained from three weeks of data with those obtained from all T_k weeks of data indicates that the parameter estimates are fairly stable in most cases, despite the few or no degrees of freedom in estimation of the model.

We draw several interesting conclusions from the adaptive forecasting scheme. First, it appears that it is generally easier to forecast, with little or no box-office data, what the cumulative box-office potential for a movie will be, as determined by the scale parameter N. However, it is significantly more difficult to predict the temporal patterns that will lead to these ultimate cumulative revenues, as determined by the shape parameters γ and λ. Further, while the meta-analysis methodology is potentially useful for forecasting box-office grosses without any data, potentially important variables of a proprietary nature (such as the production budgets, production values, promotional spending, etc.) need to be added to the list of predictor variables before the meta-analysis procedure can be reliably employed to produce box-office forecasts without any box-office data. Information on these variables would be available to an industry user of the model, and the variable lists could be suitably augmented to provide better pre-launch parameter estimates.

4. Summary, Limitations, and Future Directions

We developed a modeling framework for forecasting box-office grosses of motion pictures and tested its most parsimonious version, which we called BOXMOD-I. Our model is parsimonious, easy to understand, and easy to use. Yet its stochastic formulation offers interesting insights on the uncertainty surrounding the expected B.O. revenue patterns. Our model is a useful forecasting tool for motion picture exhibitor chains who are more concerned about later stages in the movie's life cycle, in helping them decide how long to continue exhibiting a new movie based on early box-office data. The model can also be used in conjunction with the adaptive weighing scheme discussed in §3 to generate forecasts when no data are available, and to update these forecasts as box-office data become available.

In positioning our model as a forecasting tool we made several trade-offs in favor of parsimony, which forced us to compromise on the richness of the phenomena it captures. As a result, our modeling framework has some limitations that deserve mention and can serve as directions for future refinements. In our conceptualization of the adoption process, we assumed that the time-to-decide subprocess and the time-to-act subprocess are independent. While some preliminary empirical evidence does not reject this hypothesis, it could be argued that the exhibition intensity (which influences the time-to-act parameter, and hence the supply of screens) is likely to be a function of the box-office receipts, since movie exhibitors could determine the supply (number of screens) for a new movie endogenously based on the observed demand (box-office grosses) for the movie.[6] The potential endogeneity could be addressed by an econometric model where the "supply function" of exhibitors would be expressed as box-office-receipts-dependent equation, and the "demand function" of consumers would be expressed as a supply-dependent equation. The supply and demand equations could then be estimated simultaneously as a system of equations (Judge et al. 1985). However, this econometric modeling framework would require the estimation of several additional response parameters and would not be suitable for forecasting with a few data points.

In developing BOXMOD-I, we assumed that the adoption process parameters are stationary. We outlined a relaxation of the stationarity assumption with respect to the time-to-act parameter. Stationarity in the time-to-decide parameter could also be relaxed in a related fashion. The benefit of such a richer model would, of course, have to be weighed against the cost of additional complexity and the increased data requirements.

An intriguing implementation-related direction would be to examine the extent to which it is possible to im-

[6] We thank a reviewer for bringing this possibility to our attention.

prove the predictive power of the meta-analysis procedure by using a richer set of proprietary marketing variables such as promotional spending and production budgets, which have been shown to be important determinant of box-office performance (Sawhney and Eliashberg 1995). Indeed, the opportunity to conduct such a large-scale cross-sectional analysis is unique to the motion picture industry, where data on many movies are available, and life-cycles are short enough to allow the performance of various movies released within a relatively narrow time window to be compared without the need to account for changes in macroeconomic factors and consumer tastes.

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